Muhammad Rayyan Ahmed

Ilmenau, Germany | muhammad-rayyan.ahmed@tu-ilmenau.de | +49 179 6991596 | linkedin.com/in/rayyan--ahmed
M.Sc. Data Science (in progress), TU Ilmenau | B.S. Artificial Intelligence | Four SAP Certified Associate certifications

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Cover Letter: Working Student – LLM Wiki & Agentic AI Knowledge Systems

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Dear NXP hiring team,

I am applying for the Working Student position on your LLM Wiki. You ask applicants to describe what they built: I built DocuMind (github.com/Rayyan3079/documind-rag), a RAG system that answers questions about technical documents with page-level citations, and I measured where it works and where it fails.

I am an M.Sc. Data Science student at TU Ilmenau with a B.S. in Artificial Intelligence. I use Claude and ChatGPT daily, and before my Master's I worked with business data and documentation on SAP platforms.

- **DocuMind,** I built a RAG system over PDF manuals with the Claude API that answered 91% of a 22-question test set correctly and refused all 4 unanswerable questions; hybrid BM25 and semantic retrieval raised MRR from 0.748 to 0.898
- **Failure modes,** the evaluation tested hallucination through 4 unanswerable questions, all correctly refused, and traced both wrong answers to chunk boundaries, not to the model
- **Knowledge structure,** in a team project I extracted subject-relation-object triples from text into a knowledge graph and compared graph and language-model embeddings

I would help structure knowledge so both people and AI can use it (metadata, references, templates), test retrieval and context set-ups, and check source traceability, outdated information and hallucinations with small evaluation sets. I have not yet built AI agents or worked with the Microsoft 365 integration side, and I am keen to learn both on a real production use case.

I am willing to relocate to Hamburg and can work up to 20 hours per week, starting immediately. I would be glad to show you DocuMind in an interview.

I work in English (IELTS 8.0) and am currently learning German (A1).